Research Proposal

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Thesis: The capstone is now deliberately pruned. It studies one clinically interpretable workflow: LLM-drafted pharmacogenomic or genomic medication-alert text. The evaluation asks whether a pre-display evidence gate reduces overclaiming relative to guideline-supported evidence without inappropriately denying bounded, aligned alerts.

Problem Statement

Medical LLMs can sound authoritative even when the evidence only supports a narrower statement. In pharmacogenomics this matters because a real guideline, a verified citation, a population-frequency result, or a variant assertion may support a medication-safety note without proving broad patient-outcome benefit, universal population transport, or deterministic action.

Hypothesis

Compared with ungated LLM alert text, a three-gate controller will reduce evidence overclaiming while tracking inappropriate denial. The supported claim is construct validity, not clinical utility: Stage 1 tests whether the gate enforces the intended evidence grammar on synthetic cases.

Methods

- Workflow: pharmacogenomic/genomic medication-alert text, including medication-gene pairs and variant assertions relevant to prescribing or safety review.
- Gates: endpoint/actionability, population fit, and citation/guideline support.
- Stage 1: 30 synthetic cases with expected decisions; executable controller reports overclaim detection, specificity for aligned alerts, inappropriate denial, Brier score, and error categories.
- Stage 2: replace author-designed cases with independently authored cases and blinded reviewer adjudication by qualified clinicians or pharmacogenomics reviewers.

Stage 1 Progress

Metric	Current result	Interpretation
Cases	30	Synthetic case bank for the pruned workflow.
Ungated overclaim rate	0.67	How often the draft claim is intentionally too strong.
Gated remaining overclaim	0.00	Construct-validity check, not clinical safety.
Sensitivity / specificity	1.00 / 1.00	Gate detects overclaims while allowing aligned alerts.
Inappropriate denial	0	Count of bounded aligned alerts blocked too strongly.

Value

The contribution is a submission-ready narrowing move: instead of presenting a universal assurance theory, the paper proposes a small auditable benchmark for a recurrent clinical workflow. This directly answers the critique that the prior package was broad without a single causal chain.

Selected Sources

- ML4H 2026 home. https://ml4h.ahli.cc/
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